STAPLES STEM JOURNAL



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STAPLES HIGH SCHOOL MARCH 2023

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We are thankful for the support of our excellent advisor, Ms. Amy Parent.

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The Staples STEM Journal provides an outlet for individuals to share their STEM interests with the Staples community and aims to broaden public interest and knowledge in these fields.

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Letter From the Editor

Dear Reader,
I hope you've had a wonderful start to 2023! I'm thrilled to present you with the 2023 Winter Issue, a culmination of students' hard work and efforts throughout the first semester.

This edition marks my last issue as Editor-in-Chief, and I would like to briefly reflect on what STEM Journal has meant to me. From freshman year to now, it has provided a unique space for me to grow as a communicator, explorer, and leader—to both discover new interests and cultivate others'. I am forever grateful for the mentorship I received and the empowerment I felt when I wrote my first article, along with the lessons I've since learned from leading the STEM Journal staff.

A special thank you to: Mrs. Amy Parent, who has been our faculty advisor for the past two years; and Mrs. Karen Thompson, who was our faculty advisor from the Journal's start in 2016 to 2021. Your constant guidance and support have been instrumental to our success.

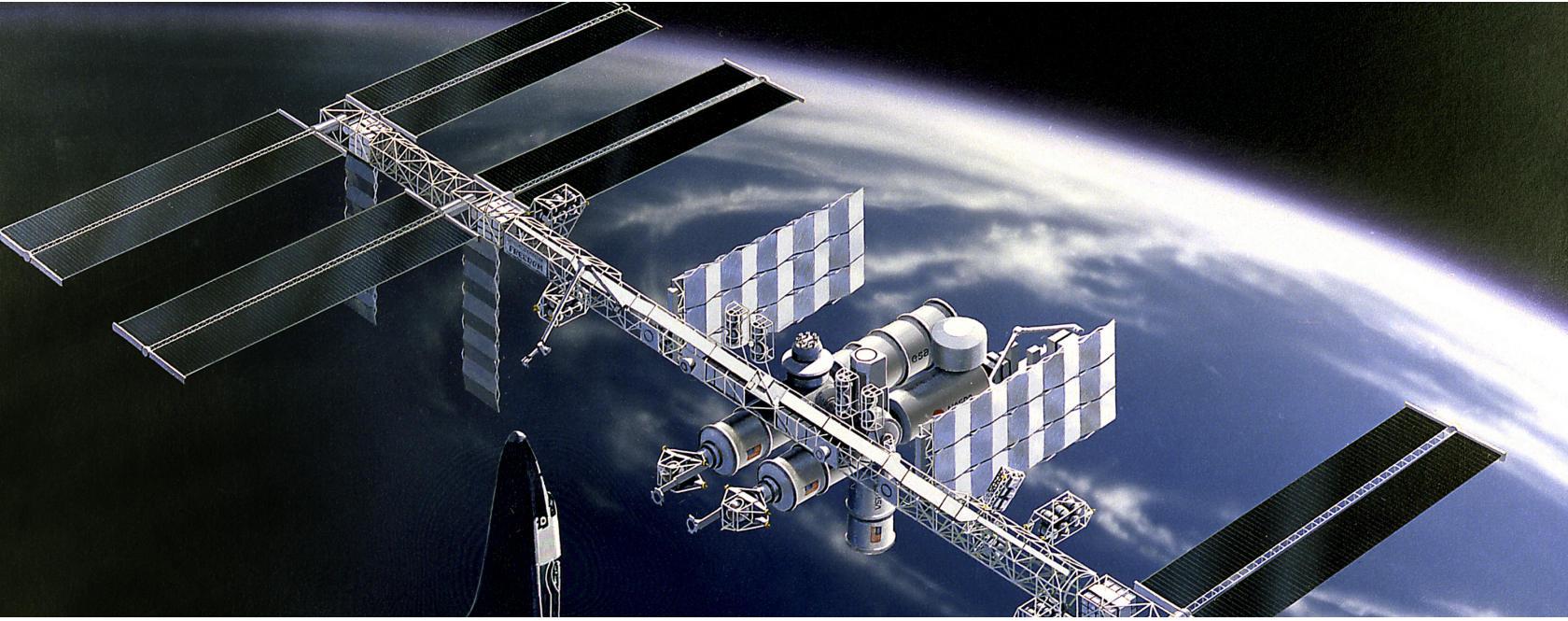
I would also like to celebrate the achievements of STEM Journal's incredible team of writers, editors, and officers. With this issue, they not only investigated critical issues surrounding environmental health and medicine, but also the fascinating histories behind the ISS and Conway's Game of Life. As always, I am in awe of their passion and drive to follow wherever their curiosities take them, and I can't wait to see all of their future accomplishments.

Today, it is more important than ever for us to promote greater communication between researchers, policymakers, and citizens. In an era of frequent distrust towards science, we must build bridges to effectively translate research into policy and foster scientific literacy; STEM writing plays an integral role in that exchange. I hope you enjoy learning about today's scientific frontiers—and emerge with a better understanding of how our world works—through students' pieces.

Finally, I extend my best wishes to next year's leadership team, including the Journal's new Editor-in-Chief, Will Boberski, and Layout Editor, Sam Zwick-Lavinsky. I'm excited to see where the Journal goes next year!

Happy reading!

Lucia Wang '23
Editor-in-Chief



1. Space Station *Freedom* and High Risk Science Megaprojects

Peter Loranger, '24

Last January, NASA released an updated version of their International Space Station Transition Report dated March 30th, 2018. With "high confidence" that it can be operated through 2030, the report states that "the ISS is now entering its third and most productive decade of utilization" (NASA, 2022). The station continues to serve its purpose as humanity's preeminent scientific lab in low Earth orbit. It furthers a variety of diverse interests—scientific, commercial, and political. Despite its current success, the station's history is complicated. The initial design and production of the station was kneecapped by a need to appease a wide coalition of special interest groups for funding and approval. This article is a retrospective review of why and how domestic politics impacted the initial development megaprojects.

The International Space Station began as Space Station *Freedom*. Announced in 1984 by President Nixon, it was considered by then-NASA administrator James Beggs and others as the "next logical step" for the American space program (Rusnak). The project initially enjoyed widespread support through its commitment to hosting a wide spectrum of experiments in "materials processing, astronomy, earth observations, and the life sciences" (Kay, 1993). However, this ambition would lead to *Freedom's* downfall.



Astronauts performing experiments aboard Space Station Freedom (Life, *1987).*

of the ISS, and how the lessons learned from its historcan be applied to future science
As the list of contributors and interested parties grew, *Freedom's* requirements ballooned to a point where the construction plans were untenable and at points contradictory. While NASA promised microgravity manufacturing facilities, docking for incoming spacecraft, and space for complicated astronomical equipment, these requirements could not all be met at the same time. The station's overly large solar panel array would create drag that would continuously pull the station into lower orbits, and the conditions required for microgravity manufacturing would be disrupted by the docking and undocking of space vehicles (Kay, 1994). To run all these systems, *Freedom* would also be required to generate 75 kW of power,



Drawing of Freedom ***living area*** *(*Crew Modules, *1987)*

approximately five times the output of the contemporary Russian *Mir* station (Congress, 1987). Additionally, an internal NASA report claimed that for the routine maintenance required to operate the intricate equipment planned for Freedom, astronauts would have to perform around 2,300 hours of extravehicular activity per year —more than six times the total extravehicular activity recorded during the entire US space program so far (Kay, 1994). The requirements for a station of such a large scale and scope as Freedom were beyond anything ever previously conceived by NASA.

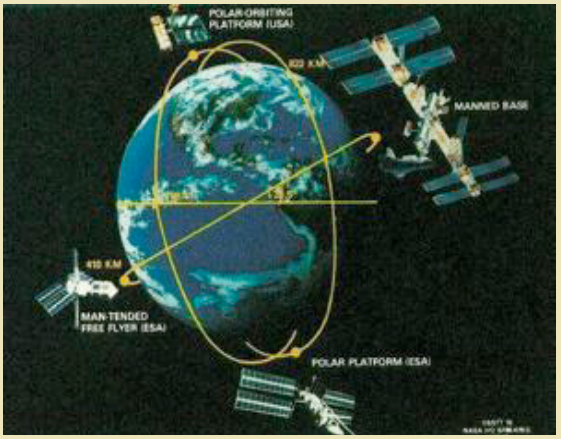
All of these contrasting requirements were too much to ask of one system. To fix these issues, the station was "redesigned into irrelevance" (Roland, 1996). Each redesign made the station smaller, less capable of accomplishing its initial goals, and more contentious. Complicating the matter was a skyrocketing projected development cost, rising from $8 billion to $37 billion between 1984 and 1990, and an immense estimated operational cost (Kay, 1994). By the early 1990s, Freedom was politically defunct, and was only saved when Russian cooperation allowed the Clinton administration to reinvent Freedom as the International Space Station.

But why did NASA, the organization with plausibly the most institutional knowledge regarding developing space programs on the planet, fail so completely when designing Space Station *Freedom*? Why did NASA feel the need to appeal to so many interest groups at once? The answer is not technological or economic, but political.

In his article "Democracy and Super Technologies", Professor W. D. Kay of Northwestern University argues that Freedom's troubled development was a result of systemic failures associated with the traditional democratic political process. According to the prevailing norms of democracy, large scale and high capital scientific investments require the ap-

proval of a broad consensus. However, the process of obtaining this consensus increases the scale and complexity of these projects and lowers their chances of success (Kay, 1994). Kay compares the Apollo program, which already had broad public support and a small set of participants, to the Freedom project, which had to bargain with a large group of potential stakeholders and lacked a base of public approval. He also argues that our democratic system "leads proponents to oversell [a] technology's capabilities while underestimating its development costs" to gain approval (Kay, 1994). In Kay's view, Freedom's failures were an unavoidable result of ingrained democratic and populist pressure.

However, Professor Alex Roland of Duke University posits a different explanation: the phenomenon of buy-in, the "now-venerable Washington technique of intentionally underestimating a program's cost and overestimating its payoff" (Roland, 1996). Roland argues that NASA deliberately promised Congress impossible results to sell Space Station F*reedom,* and he emphasizes the importance of considering technological and economic limitations when developing risky scientific projects. In his view, instead of Freedom's failures being the fault of inevitable systemic issues, the blame lies with NASA, who "sacrificed technical and economic judgment on the altar of politics" (Roland, 1996).



Plan for network of space platforms and satelittes related to Freedom (Baseline, *1987)*

One possible avenue is reforming the systems that encourage deceitful behavior. Bent Flyvbjerg, a professor at the University of Oxford and expert on megaproject management, advocates for increasing the legal and managerial responsibilities megaprojects have to their shareholders and the general public. In his article, "Design by Deception: The Politics of Megaproject Approval", he recommends furthering "democratic governance" over risky megaprojects (Flyvbjerg, 2005). While this may seem contradictory, as one of the contributing motivations to Freedom's buy-in and feature creep was having to seek approval from a wide consensus of stakeholders, it is a proven strategy. The International Space Station succeeded where Space Station Freedom failed in part because it was supported by an international coalition of interested nations: Russia, the United States, Canada, Japan, and the European Union. Making the scientific organizations behind future megaprojects more accountable will help restrict their incentive to misrepresent the benefits and risks of a project. The United Kingdom began implementing a similar accountability requirement in 2003, making funding "unavailable for projects that do not take into account this [buy-in] bias" (Flyvbjerg, 2005). On the other hand, Kay takes a more deregulatory stance. He argues that similar to how megaprojects are developed primarily by governments because of their extraordinary economic demands, "eliminating the political problems that surround such

development would require extraordinary governmental measures insulting these technologies from traditional political pressures" (Kay, 1994). He sees the solution to the failure of megaprojects in decreasing the pool of potential stakeholders and decreasing the requirements for a project getting governmental approval.

The dichotomy of these two positions —one advocating for an increase in democratic oversight and the other arguing for an undemocratic and technocratic system— underscores the political tension inherent to developing high risk scientific megaprojects, a conflict that will only increase as dependence on governments to fund scientific breakthroughs grows.

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2. Alzheimer's Disease: A Modern Threat with Modern Solutions

Abraham Lobsenz, '25

Alzhemimer's is the most common neurodegenerative disease in the world (along with Parkinson's), affecting as many as 6.2 million people in 2022 according to the Alzheimer's Disease (AD) Association (Hollander, 2022). One wonders how such a prevalent disease could have no remedy and generally ineffective treatment protocol. Indeed, cultivating treatment options is imperative in a future of seemingly elevated neurodegenerative stress. Of all of the challenges in dealing with neurodegenerative diseases, the foremost is deciphering what factors play into its manifestation. In the case of Alzheimers, a fundamental vagueness in scientists' understanding of the brain and its various subparts makes this extremely difficult. So, given its flaws, how does science comprehend Alzheimer's today?

Control vs. Alzheimer's

In a normally functioning brain, the basic structure of a neuron is as follows: a synaptic connection through a "commutative" axon and several "receiving" dendrites. Essentially, an electrical impulse travels down the length of the axon, promoting the release of neurotransmitters across a gap (synapse) into the dendrites of surrounding neurons. This neurotransmitter signaling is the basis for your ability to read this sentence. These axons and dendrites constantly restructure and remodel themselves depending on the stimulus received from other neurons, helping to maintain and develop the synaptic connections of these cells, which are expected to live up to 100 years. Herein lies the difference between neuronal and other somatic cells, being that neurons can't self-replicate and are thus subject to a final death. The main cell body consists of the genetic information and the proteins expressed by it, and also serves as a regulatory center. Neurons are energy demanding, and consume 20% of energy used by the human body, meaning constant access to glucose

and oxygen for cellular respiration is a necessity for the survival of neurons (NIA, 2017). As one ages, they are naturally subject to both holistic and local decay. The brain may naturally sacrifice some volume (~2.2g-2.7g) and synaptic connections may be interrupted as that 100-year expiration date nears, but not to the extent of even preliminary AD (Hartmann, 1994). AD typically spawns in the hippocampus and entorhinal cortex (Figure 1), which compose the temporal lobe, appropriately named for its memory-associated function. For this reason, memory lapses are common in early AD. Proximally, it will affect areas of the cerebral cortex responsible for language, reasoning, and social behavior. Alzheimer's spreads throughout the entire brain, before a patient loses all ability to function alone, quickly followed by brain failure and fatality (NIA, 2017). On the cellular level, this appears like neurons abandoning their basic physiological functions, refraining from supporting synaptic connections and forfeiting their hyper-metabolic function.

Factors

In the attempt to discover the factors associated with Alzheimer's, one enters a more precarious scientific domain. While some contributors are well known, such as Tau accumulation and hyperphosphorylation, and Beta-Amyloid plaques, there are a host of additional causal relationships in AD, as well as an incomplete description of how these multiple factors may interact with each other. What is known is this: Tau proteins, which line nutrient-transporting microfibrils in healthy neurons, act erratically in AD. They detach from their respective microfibrils and form string-like structures of their own, called neurofibrillary tangles. These tangles inhibit nutrient-transportation to axons and dendrites and therefore impair synaptic connection between neurons (NIA, 2017). This Tau is also hyper phosphorylated, a characteristic that results in axon shortening. Beta-Amyloid plaques in interneuron space are another well-known factor of AD. They can be traced back to Amyloid precursor protein, which subsequently decays into B-Amyloid. While this protein is normally involved in the communication between neurons, it is produced in abnormal amounts during AD progression, forming plaques that in turn disrupt cell function. As an example of some uncertainty in this field, novel research has recognized that there must be some interplay between Tau and B-amyloid, but it is not definitely understood (NIA, 2017). Yet another poorly-understood factor lies in the disrupted function

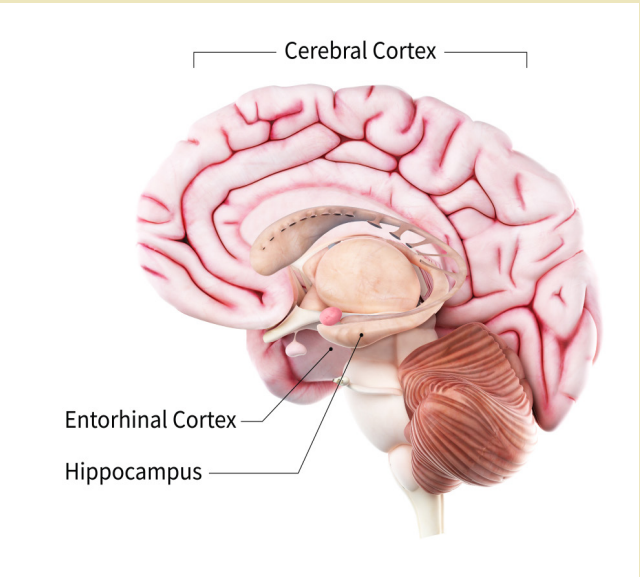


Figure 1: A model of the Cerebral Cortex, Entorhinal Cortex, and Hippocampus (NIA, 2017)

of glial cells. Glial cells can be thought of as the chaperones of various somatic systems. In the central nervous system, microglial cells are especially important. They normally clean the interneuron space of harmful waste, but in Alzheimers they fail to respond to B-Amyloid plaques. They arrive at the neuron, but ignore the waste in front of them. Said failure is being investigated and linked to gene loci, such as TREM2 or NFKB (NIA, 2017)(Tan, 2008). Finally, vascular problems have both a cause and effect correlation with AD. Vascular stress deteriorates the Blood Brain Barrier (BBB) over time, preventing it from accomplishing its function of allowing entry to crucial nutrients, and denying it to harmful agents. AD also leads to vascular problems as B-Amyloid plaques in brain arteries cause blockages, exacerbating the issue (NIA, 2017). As can be seen, AD is a mess of complex interactions between already questionably understood phenomena.

Potential Treatment Options

AD's intricacy has not deterred scientists, who are focusing on several promising treatment candidates, of which two will be explored here. Firstly, the manipulation of the sigma-1 receptor (S1R) of the endoplasmic reticulum is a potential option, because of its advantageous position already within neurons, and its all-around positive implications in its metabolic pathway. One such implication is its crucial role in calcium balance in the mitochondrial membrane. A study found that S1R bound with myristic acid (MA) promoted the degradation of p35, a necessary component for p25, which is the activator of an enzyme, cyclin-dependent kinase 5 (cdk5) (Tsai, 2015). This enzyme is usually overproduced in tauopathy (Tau accumulation). The p35 activator is unstable, and is quickly converted to p25 by a calpain-mediated cleaving process. This more stable compound promotes the abnormal production of hyperphosphorylated Tau, which shortens the axon and generates neurofibrillary tangles. S1R may bind with MA, which in turn causes p35 degradation before having an opportunity to convert into p25. This argument is supported by the study's analysis of S1R-deficient neurons, in which p35 appears to be over-abundant, providing more substrate for the production of p25 and thus cdk5. To summarize, S1R activation and MA exposure could stunt Tau hyperphosphorylation, preventing axon shortening, and could also reduce the activation of cdk5 in AD. Another study helped to confirm the relationship of S1R in neurodegenerative diseases by means of an in vivo (patient-based) study, establishing that early AD patients did indeed have a lower density of the receptors (Mishina, 2008).

Another promising candidate is Epigallocatechin gallate (EGCG), a compound with special biochemical properties found in green tea. One study aimed to demonstrate the application of EGCG in AD treatment (Seidler, 2022). While the EGCG is not very bioavailable, nor is it stable enough in the bloodstream to pass the BBB and make it to Alzheimer's implicated neurons, in vitro models of its contact with tau fibrils have proven impressive in treatment. As can be seen in Figure 2, brain extract incubated with EGCG promoted Tau disaggregation, and those fibrils that did remain after 24 hours became swollen and unstable. This effect is due to EGCG's structural nature, which embeds between Tau proteins in a fibril and

interacts with polar hydrogen bonds in such a way as to "rip apart" the Tau fibrils, as illustrated in figure d and f. EGCG may also have a positive effect on B-amyloid plaques.

AD is an extremely challenging case, but one undergoing significant treatment research, and one with hope of a resolution in the near future. It is important to note that the research involved with AD is not only critical to those 6.2 million Americans who struggle with it, and the more who will in the future. Alzheimer's represents a major point in the frontier of neurology, for the thousands of studies on AD and other neurodegenerative diseases have begun to shed light on the complexities of the brain. AD research is pioneering humanity's understanding of themselves, and the weird 1.25 kg lump in their heads. As with vaccines a century ago, nothing is more certain than science pushing its limits, no matter how confining those boundaries may seem at the moment.

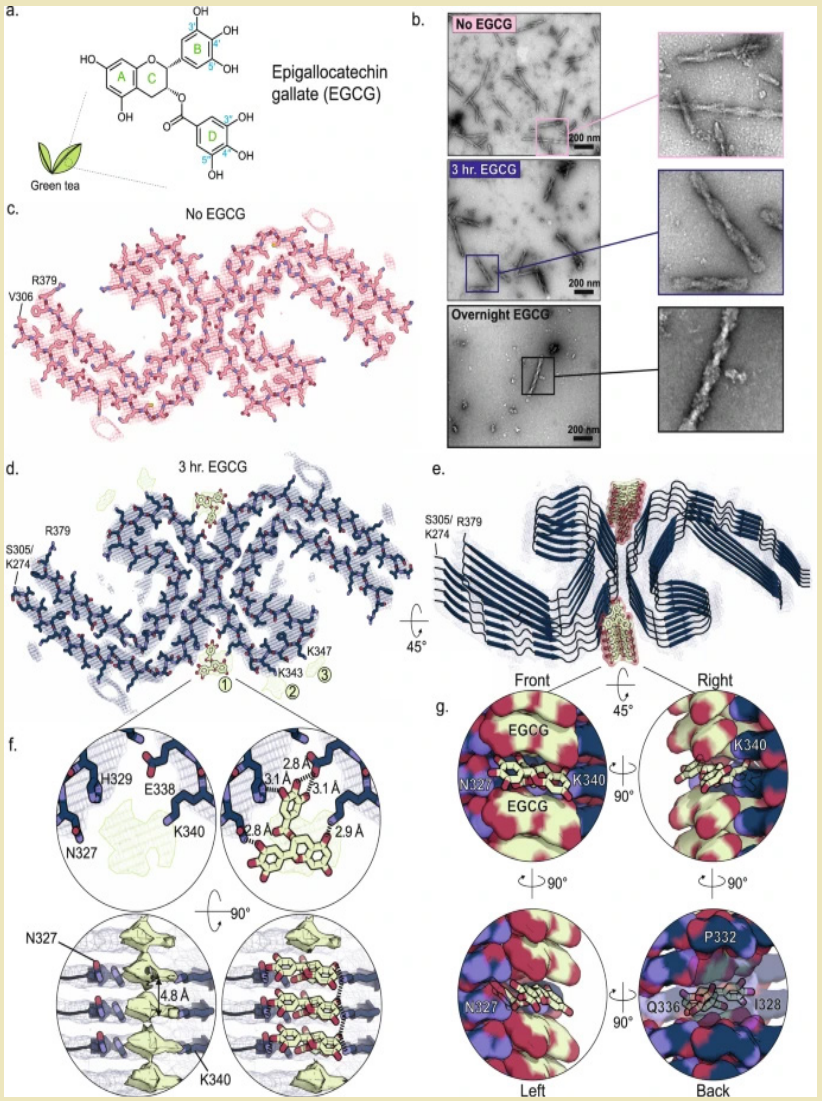


Figure B: Diagrams and Imaging of EGCG involvement in Tau Disaggregation (Seidler, 2022)

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3. Impulsivity: A Controllable Setback

Tishia Darmawan, ‘24

Everyone makes decisions on a whim, it’s simply human nature. Impulses come suddenly and strongly in an attempt to persuade the body to begin an action while thoughts tend to trickle in more slowly and with more rationale behind them. While some impulsive choices can be positive and even necessary for survival, such as breathing and blinking, others can be negative, like consuming unhealthy foods or participating in irresponsible dares (Caroli, 2015). Regardless, they happen. People naturally succumb to their most basic desires and instantaneous whims. Whether a good idea or a bad one, impulses must first travel through the brain’s prefrontal cortex, a place where these fast paced decisions are synthesized, approved and even quashed (Mitchell & Potenza, 2014).

The Prefrontal Cortex

A vast region of the brain beginning just behind the forehead, the prefrontal cortex and its corresponding sub-regions formulate and execute high-level thoughts, voluntary motor actions and sustained attention, as well as inhibitory control and stimulus detection. Often known because of the infamous Phineas Gage, a man who had a pointed iron bar blasted through his skull during an explosion but miraculously lived, the prefrontal cortex plays a significant role in personality and behavioral patterns (Siddiqui, 2008). Following his horrific accident, Gage was never the same. While prior to the incident he was considered a model employee, he was never able to regain quite the same level of professionalism or really any shadow of his former self. He used offensive profanity, exhibited indifference to his peers and previous friends, and seemed to have no sense of balance between his instincts and thoughts (Twomey, 2010). Essentially, he was unable to control his impulses and was incapable of experiencing high levels of both intellectual thought and physical function due to the severe damage to his prefrontal cortex.

The Pomodoro Technique

While Italians may be familiar with 'pomodoro' as 'tomato', the Pomodoro Technique is actually a time management method that's easy to pick up and simple to use in order to be more efficient with the time available and productive with the possessed skills. Developed by Francesco Cirillo in the late 1980s, the Pomodoro Technique is used by people worldwide in order to stay focused and get more done in short periods of time. Frustrated by his lack of focus, he questioned if he could even maintain a working state for 2 consecutive minutes. To put this challenge to the test, he went to his kitchen and found a tomato-themed timer. Thankfully, he was able to succeed in his 2-minute dare, and as a result tested the method for its ideal parameters.

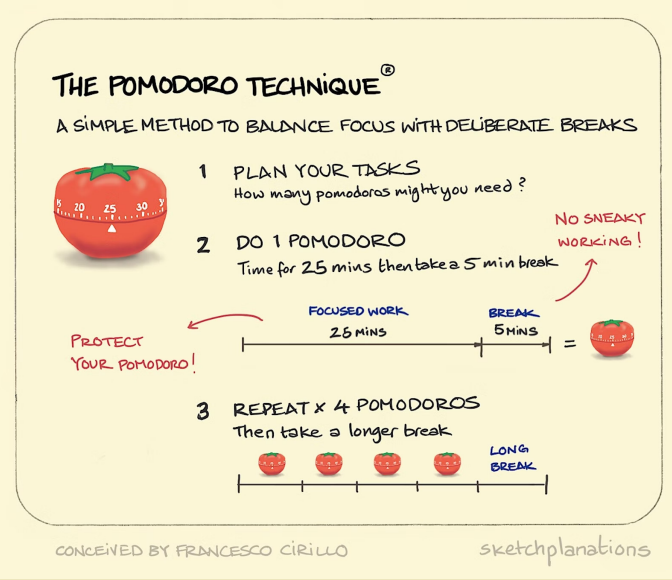


Figure 1: A diagram explaining the Pomodoro Technique ("The Pomodoro Technique", 2022)

What he came up with is simple: a technique named after his timer that blocks together sessions of focus and rest to help himself and people all over the globe get more done. To be specific, his method consists of selecting a task or project to focus on, attaining a timer and setting it for 25 minutes, working through this block and ignoring any distractions or intruding thoughts as best as possible, and finally upon completion taking a 2-5 minute break and noting the successful work block with a check or 'X' in a notebook. After the break, repeat this process until all the work has been completed or all the available free time has elapsed. A very important step to take after working hard for four sessions is taking a longer break to mentally reset and get ready for another session. After this extended break, feel free to switch tasks or projects and begin working on something new or continue working on the previous one to finish it off. The choice is yours. Essential notes for success however, include not working through the scheduled break and not getting distracted

during the scheduled working period. While the Pomodoro Technique is designed to help its user stay on task, the end result will always depend on the user's choices. (The Pomodoro® Technique | Cirillo Consulting GmbH, n.d.).

Why Does it Work?

When trying to be productive, impulsive decisions and intrusive thoughts are often the cause of a setback. The human mind naturally tends to wander, and stopping this impulse keeps people productive and focused. By placing the mind under a time limit, the brain is instilled with a sense of urgency. It essentially reminds the brain that it has only a certain amount of time to get as much work done as possible. This is only possible because the pre-frontal cortex is heavily connected to the amygdala. The amygdala is responsible for many emotional choices as well as choices that involve a sense of emergency. Overall, the amygdala kickstarts the brain's fight or flight response and allows it to focus on the task at hand instead of letting the mind wander to other thoughts and pleasures. In addition, the promise of periodical rest after the completion of a certain amount of work aids the brain in making personal decisions regarding staying on task and maintaining focus. Knowing that there will be time to do whatever was forgotten or whatever may seem more interesting and enticing allows the brain to override any instantaneous and impulsive choices (Boogaard, 2022).

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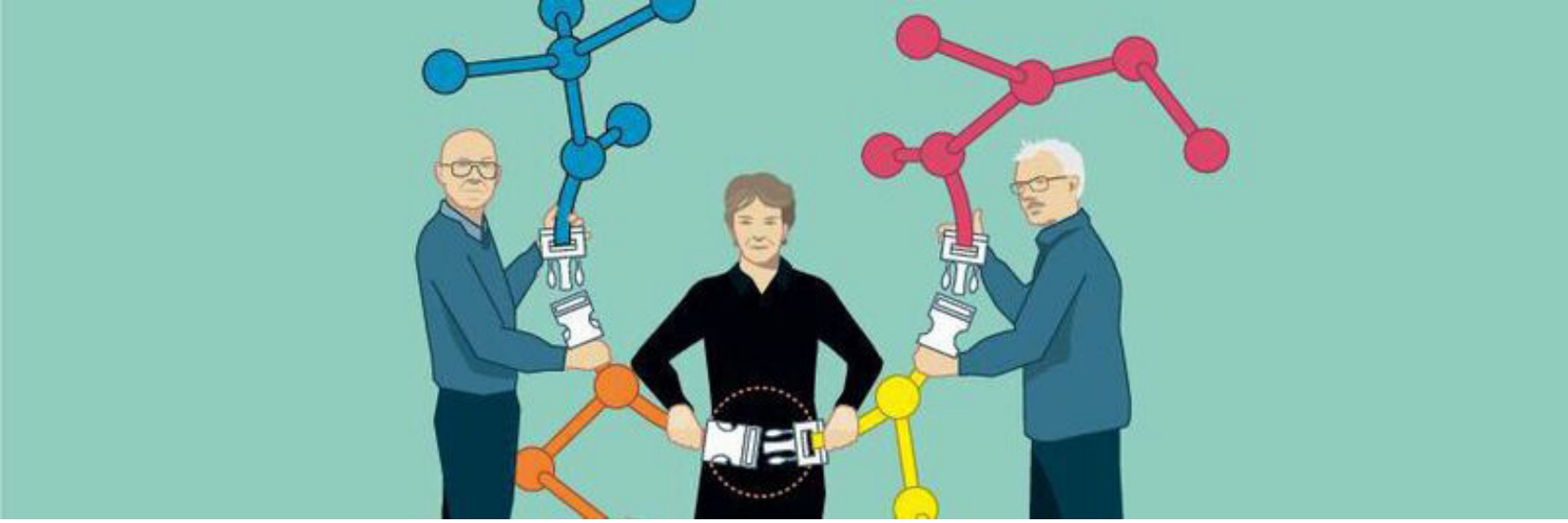
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4. Click!: How Molecular Building Blocks Are Changing Medicine, Chemistry and Biology

Will Boberski, ‘25

Our world is made of complex organic molecules; yet, chemists have historically struggled to efficiently synthesize such molecules, often using time consuming processes and creating unwanted by-products. This year’s Nobel Prize in Chemistry honors chemists who have brought the simplicity and functionality we take for granted in the physical world to the chemical world. The future of medicine and organic chemistry may lie in click chemistry,

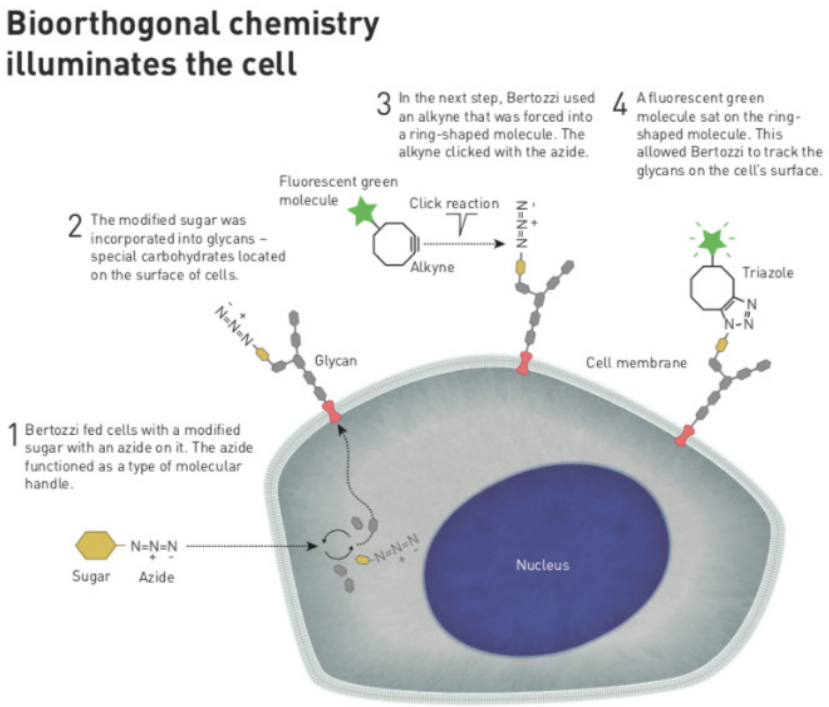


Figure 3. Bertozzi used bioorthogonal chemistry to add a chemical tag (a fluorescent green molecule) to the glycans she was studying (Royal Swedish Academy of Sciences, 2022).

the molecular equivalent of playing with Lego blocks.

Click Chemistry and the CuAAC Reaction

The 2022 Nobel Prize in Chemistry was awarded to K. Barry Sharpless (United States) and Morten Meldal (Denmark) for the development of click chemistry, as well as Carolyn Bertozzi (United States) for the development of bioorthogonal chemistry (Fernholm, 2022). In a 2001 paper, Sharpless, who also won the prize in 2001, proposed that "organic synthesis conducted as it has been, in imitation of nature's carbonyl chemistry [chemistry involving compounds with a carbonyl group (Encyclopaedia Britannica, "Carbonyl group," 2018)], is ill suited for the rapid discovery of new molecules with desired properties" (Kolb, 2001). Sharpless and his fellow click chemistry innovators believe that the future of chemistry doesn't lie in mimicking natural processes, which can be complex, inefficient, and generate unwanted byproducts. Instead, linking atoms other than carbon or hydrogen, known as heteroatoms (Encyclopaedia Britannica, "Chemical compound," 2022), can help "develop an expanding set of powerful, selective, and modular 'blocks' that work reliably in both small- and large-scale applications." Sharpless defines a click reaction as modular, wide in scope, high yield, generating benign byproducts that can be easily removed, and having simple reaction conditions (Kolb, 2001).

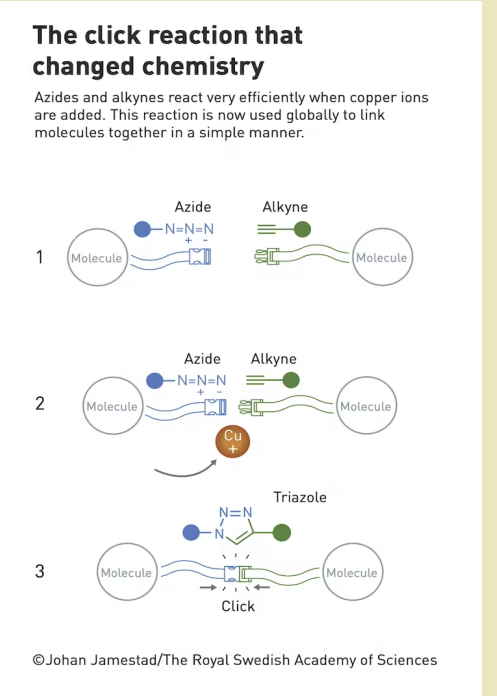


Figure 1. The CuAAC reaction "clicks" together azides and alkynes by adding copper ions as a catalyst (Royal Swedish Academy of Sciences, 2022).

The fundamental reaction of click chemistry—copper catalyzed azide-alkyne cycloaddition (CuAAC) (Scripps Research, 2022)—was discovered by accident. After reacting an alkyne with an acyl halide, a routine organic chemistry reaction catalyzed by copper ions, Morten Meldal and his colleagues realized that their alkyne had reacted with the wrong end of the acyl halide molecule. The azide had bonded to the alkyne, creating a ring-shaped compound known as a triazole; the copper ions had controlled the reaction so that the acyl halide had remained untouched. At around the same time, Sharpless independently published a paper describing the CuAAC reaction as click chemistry, since it met the requirements outlined in his 2001 paper (Fernholm, 2022). Click chemistry, especially the CuAAC reaction, has a wide range of potential applications in medicine, manufacturing and biotechnology.

The CuAAC reaction has an enormous range of potential applications, since it enables the creation of molecular Lego blocks that can easily be attached to the ends of other molecules. A manufacturer of plastics or fibers could "click in substances that conduct electricity, capture sunlight, are antibacterial [or] protect from ultraviolet radiation," and a pharmaceutical company could cheaply and efficiently assemble potential drugs (Fernholm, 2022). Sharpless's lab has since discovered other click reactions, like sulfur-fluoride exchange (SuFEx) (Finn, 2022) and the thiol-ene reaction (Scripps Research, 2022). The discovery of multiple click reactions increases the applications, and thus the utility, of click chemistry.

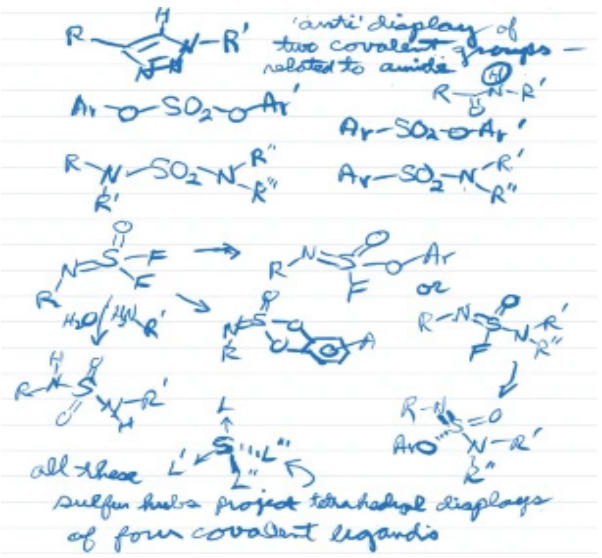


Figure 2. Sharpless used notebooks to sketch out his ideas, including this reaction with triazole and sulfur (VI) connectors (Finn, 2022).

Bioorthogonal Chemistry

The field of bioorthogonal chemistry began in the 1990s, when Carolyn Bertozzi embarked on a project to map a glycan that draws immune cells towards lymph nodes. Glycans are complex carbohydrates often found on the surfaces of cells, but molecular biology lacked efficient tools for tracking them. When Bertozzi heard that a German scientist had prouced a variant of sialic acid (a sugar component of glycans) *in vivo*, or within living cells, she wondered whether her cells could create a modified type of sialic acid (for example, with an added fluorescent molecule) to map and track glycans. For the reaction to be effective, Bertozzi's chemical attachment had to be insensitive to everything else in the cell; eventually, she found an azide that could meet her criteria, and modified it using the Staudinger reaction to add a fluorescent molecule (Fernholm, 2022). Bertozzi called her new approach bioorthogonal chemistry, or "chemical reactions that neither interfere with nor interact with biological systems" (Palmer, 2010). The introduction of these azides into living mice and zebrafish enabled the first imaging of glycans in living cells (National Inventors Hall of Fame, 2017). According to a primer on bioorthogonal chemistry published in *Nature*, "Bioorthogonal chemistry has significant overlap with the broader field of click chemistry, which is defined by high-yielding and modular reactions that are wide in scope, simple to perform and generate only inert by-products" (Scinto, 2021). Although Bertozzi realized that her azide could be used in click chemistry reactions, she also knew that copper was toxic to cells. Instead of using copper ions, she forced the alkyne in a ring to create potential energy. This energy could then be used to catalyze the reaction, a process she termed strain-promoted alkyne-azide cycloaddition (Fernholm, 2022).

Bioorthogonal chemistry has been used to map internal structures, track the movements

of cells and tumors, sequence DNA, and study viruses (Engelbrecht, 2022). When Bertozzi and her team discovered that some glycans appear to protect tumors from the immune system, they joined a glycan antibody with enzymes that break down glycans on tumor cells (Fernholm, 2022). Bertozzi and her colleagues named this technology SMARTag, and formed Redwood Biosciences to commercialize their discovery (National Inventors Hall of Fame, 2017). Bertozzi's technology can also target specific cells based on changes in cell surface sugars that indicate cancerous cells or malignancy (Palmer, 2010).

The 2022 Nobel Prize in Chemistry winners share a vision for a more functional, efficient future in chemistry and medicine. As Sharpless stated in a paper in *Nature Synthesis*, "A day will come when most biological targets—proteins, nucleic acids, cells, tissues—will be addressable by specific [pharmaceuticals] of their own, developed quickly under natural conditions using ligation reactivity that is sensitive to the molecular environment in need" (Finn, 2022). With just a few clicks, entire fields of organic chemistry have been unlocked, waiting to be explored.

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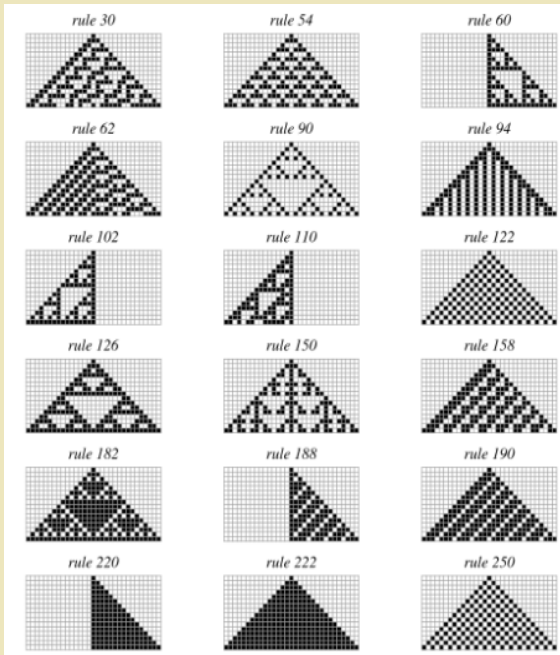
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5. The Legacy of Conway's Game of Life

Zachary Gottlieb, '25

A thousand cells are propagating across an endless grid, evolving and devolving into a seemingly random pattern. But Conway's Game of Life isn't pure chaos; these "organisms" are actually governed by a simple set of laws that work together to produce a beautifully enigmatic and astonishingly diverse simulation. This is the "GoL", or simply, "Life", the awkward brainchild of English mathematician John Horton Conway. Though it was created over 50 years ago in 1970, when Conway published his game in Martin Gardner's Mathematical Games section of *Scientific American,* Life has drawn a cult following, devoted to the simplistic beauty of Conway's Game, which demonstrates the principles of Deterministic Chaos and the Hierarchy of Life (Roberts, 2020).



Elementary Cellular Automata from Wolfram's "A New Kind of Science" (Weisstein)

What is a Cellular Automaton?

First theorized by the mathematicians John von Neumann and Stanislaw Ulam, whom Conway cites as an inspiration for his game, cellular automaton (plural, automata) is a computational model of a set of colored cells on a grid (Emmite et al.). Over iterative generations, the states of the cells are updated based on the surrounding states of the neighboring cells (Weisstein). Specifications

such as grid shape, the number of cell states (essentially colors; often expressed as k = 2), and the "neighborhood" which is the size of the surrounding cells that can affect the central cell - that is generally either a Moore neighborhood (square; 8 cells) or the von Neumann neighborhood (diamond; 4 cells). Based on these factors the composition of the grid evolves into an intricate pattern as the simulation progresses.

Since then, cellular automata and its properties have been used in a variety of research, applied to physical, biological and other such models, to study fields ranging from weather forecasts to simulating galaxies. One such model is the Lattice Gas Cellular Automaton, which is used to simulate the flow of fluids (Wolfram, 2002). Later, in 2002, Stephen Wolfram published *A New Kind of Science*, giving insight into his comprehensive studies of cellular automata.

Game of Life - A How to Play

Conway's Game of Life is the most well known model of cellular automata. GoL is set on an endless 2d grid, with k = 2, and a Moore neighborhood at a range of 1. The player turns on any configuration of cells, and through each successive generation, that cell and its surrounding cells are turned off and on (this is known as being "dead" or "live"). There are only 3 basic rules the game follows, and from iteration to iteration, as the game advances, these 3 rules dictate how the system plays out (Gardner, 1970):

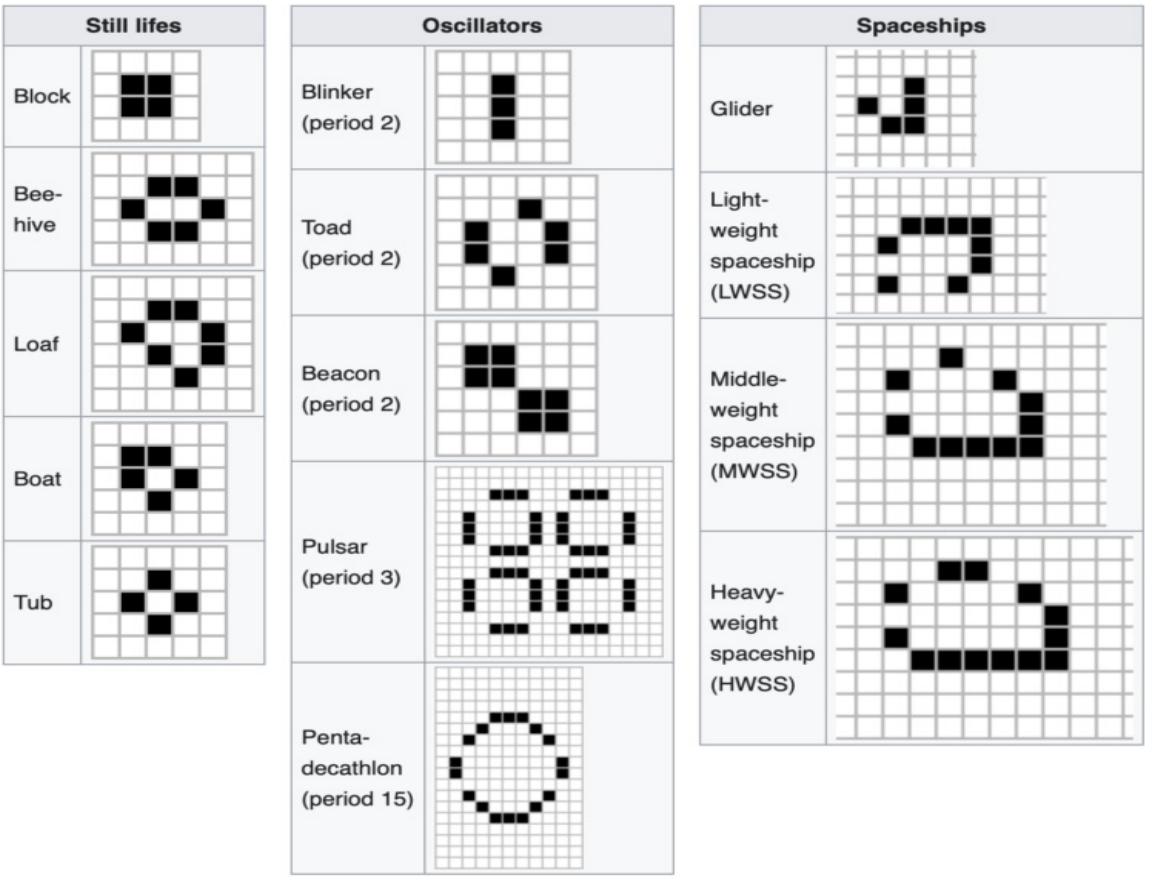
1. *Death*: A live cell with less than 2 or greater than 3 living neighbors dies
2. *Survival*: A live cell with 2 or 3 neighbors remains live
3. *Birth*: A dead cell with 3 living neighbors becomes live

Life Forms and Discoveries

When Conway originally experimented with GoL it was a long time before the invention of the computer. Thus, hunting for patterns and "life forms" - as you can imagine - took a long time. With the development and popularization of computers, GoL attracted many followers who sought to find new undiscovered life (which are recurring patterns that interact with the environment).

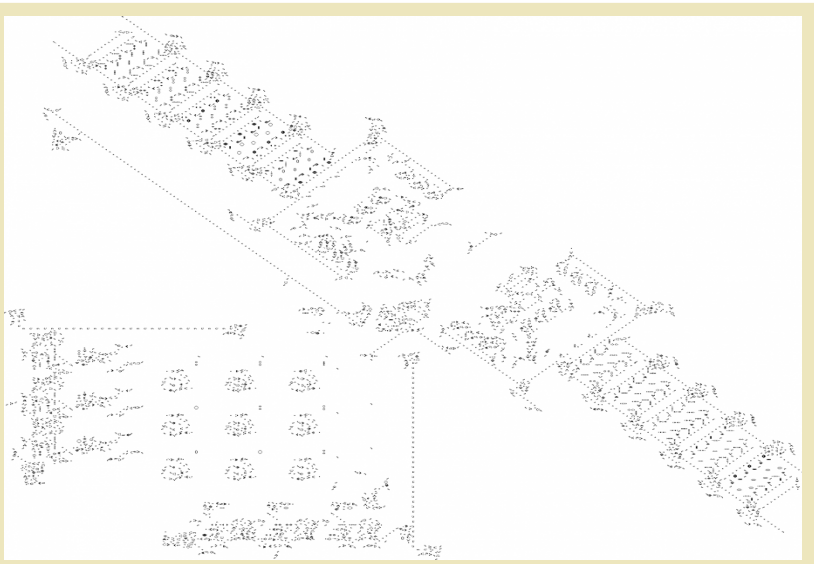
Conway categorized the discoveries into 3 distinct categories (though there are some life forms that fit into multiple classifications): Still Lifes, Oscillators, and Spaceships.

Still Lifes: (e.g. bee-hive) Stable patterns that once established, don't change over generations.
Oscillators: (e.g. blinker) Repeating patterns over successive generations.
Spaceships: (e.g. glider) Patterns that form in a different location after generation(s).



Three life form classifications with examples (Lefebvre, 2022)

To Conway, spaceships were the most interesting class of life. In larger, more elaborate simulations, these spaceships are key to transmitting information across the screen. The Glider Gun was one of the first (and smallest) spaceships to be discovered, and today some are over a hundred cells large. Though there are only 3 rules, GoL is deceiving; It is anything but a simple game. Vast megastructures have been constructed with thousands of cells, with complex interactions and exchanges of information. GoL has been proven to be a universal cellular automaton, meaning that it is capable of simulating a Turing Machine, so theoretically GoL is able to



Turing Machine in Game of Life (Johnston, 2009)

do anything a computer is capable of (De Mol, 2018). Experimentation with GoL has gone so far to construct a digital clock and even GoL inside GoL (search it up, it's pretty cool).

Game of Life and Deterministic Chaos

The patterns and cell formations produced in GoL appear to be erratic, but they're just a product of the initial structure in cooperation with the rules. This is exactly what the field of Chaos Theory is concerned with: The study of apparently random or unpredictable behaviors in a system governed by deterministic laws (Bishop, 2017). Deterministic chaos suggests a connection between the familiar and the chaotic that is often regarded as incompatible. Of course, this is a rather crude simplification, but there is an undeniable correlation in concept.

It's important to understand that Chaos Theory deals with the nonlinear, the effectively impossible to predict, such as turbulence, weather, etc. In GoL a small arrangement can generate an seemingly unpredictable system with chaotic behavior. Thus, it makes sense that cellular automata have likewise been used to simulate real-world processes. GoL demonstrates one of the core Chaos Theory principles: The Butterfly Effect. Small changes in the initial conditions lead to drastic changes in the results (Wolfram, 2002).

Game of Life and Hierarchy of Life

GoL is intrinsically alike to the real world, reflecting the processes of biology. After all, it is a game of *life* bringing out "analogies with the rise, fall and alternations of a society of living organisms" (Gardner, 1970). GoL mirrors the hierarchy of life in a computer simulation, with cells, organisms, and communities working together. The operation of the 3 rules produces emergent new forms and properties, at first a rather humble model, that matures into a labyrinthine structure. In GoL, cells conjoin to form life forms, in some intricate simulations those life forms transfer information and work towards a common function, in biology this is comparable to an organ system.

The interactions between the cells have similar biological counterparts: Cell division, cell movement, cell adhesion, differentiation, induction, and cell death (Caballero, Hodge, Hernandez, 2016). Cell death is exhibited by the death rule, wherein, the ecological concepts of overpopulation and isolation are at play.

Conclusion

Conway died in April of 2020, the 50th anniversary of the publication of GoL of which he called "a fantastic solitaire passtime" (Gardner, 1970). His game has become somewhat of a cultural wonder in the scientific community, introducing millions of people into the world of cellular automata. Since then, GoL has taught the importance of sensitivity to initial conditions and complexity arises from simplicity.

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6. Neuromodulation and Gene Therapy Prove Beneficial for Treating Neurological Disorders

Sarah Morran, '24

In the last century, there have been remarkable advancements in medical treatments and cures for various diseases. A notable example being cardiovascular diseases, such as atherosclerotic cardiovascular disease, or clogged arteries, in which people are able to receive open heart surgery. Despite the advancements in various domains of medicine, central nervous system diseases have proven themselves the exceptions. These include Alzheimer's, Parkingson's, Huntington's, Amyotrophic Lateral Sclerosis, and others. Scientists continue to study the central nervous system and have made attempts to develop medications to decelerate the progression of these diseases, or cure it entirely. Contrasting the numerous failures that new drugs have faced, there are promising new treatments that are in development.

Alzheimer's Disease

Alzheimer's disease is the most common type of dementia, and has affected an estimated 5.8 million Americans aged 65 or older in 2020. Alzheimer's disease is a progressive disease that initially affects memory loss and leads to being unable to respond to the environment (Centers for Disease Control and Prevention, n.d.). Alzheimer's research was guided by the idea that Alzheimer's was caused by the beta-amyloid protein, or plaques, which are indicators of Alzheimer's disease. Therefore, Alzeimer's treatments were developed to identify and remove microscopic clumps of the beta-amyloid protein (Mayo Clinic Staff, n.d.). In June 2021, the U.S. Food and Drug Administration (FDA) approved Aduhelm (aducanumab) for the treatment of Alzheimer's disease. Differing from the ten-month standard time needed to approve new drugs, Aduhelm was approved through the "accelerated approval pathway," cutting the approval time to six months. This accelerated approval process is

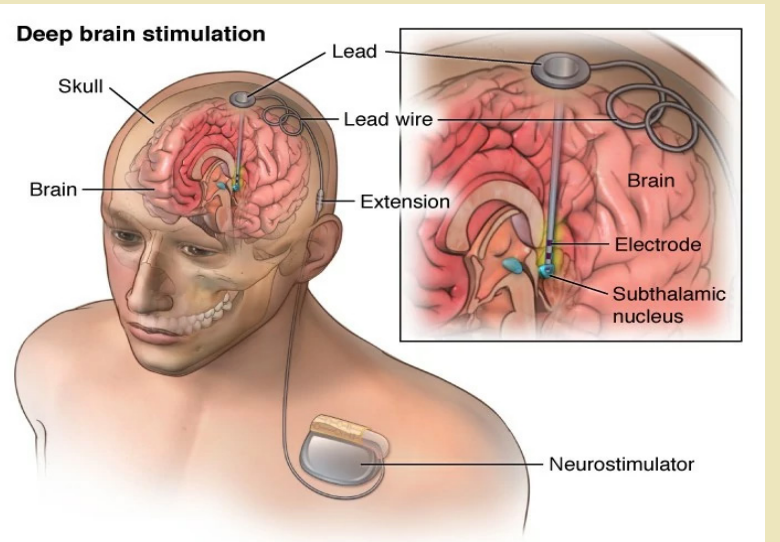
designed to allow "drugs for serious conditions that filled an unmet medical need to be approved based on a surrogate endpoint" ("FDA Grants Accelerated Approval for Alzheimer's Drug", 2021). This enabled the FDA to approve drugs faster, and evaluate their effectiveness at a later point. Despite its rapid approval, the evaluation of Aduhelm by the FDA stated that the data was inconclusive as there were several issues with its trial data. Additionally, the studies that were done for Aduhelm were stopped early and were incomplete (Tampi et al., 2021). Furthermore, some parts of the study were inconsistent, as there was occasional unblinding for dose management. These factors caused the Aduhelm medication to be unreliable.

Despite this notion of a direct relationship between plaques and Alzheimer's, Alberto Espay, professor of neurology in the UC College of Medicine, states, "The paradox is that so many of us accrue plaques in our brains as we age, and yet so few of us with plaques go on to develop dementia" (Tedeschi, 2022). This new insight also contributes to the failures in developing Alzhemer's treatments.

New Approaches

Despite the shortcomings that emerge in the development of treatments for neurological disorders, there is a revived hope that new strategies will be effective at treating these disorders. These new approaches to treatment include neuroimmunology, neuromodulation, gene editing, stem-cell transplants, RNA therapies, and more.

Neuromodulation alters the nerve activity through targeted delivery of a stimulant to specific neurological areas in the body. Neuromodulation incorporates deep brain stimulation, vagal nerve, or the main nerves of the parasympathetic nervous system, stimulation, and transcranial magnetic and electrical stimulation (Lim et al., n.d.). transcranial magnetic and electrical stimulation (Lim et al., n.d.). This approach will be able to relieve people suffering from certain neurological disorders, such as restoring function to areas of the body or alleviating pain (*Neuromodulation Therapies - Patient Information*, n.d.). Deep brain stimulation (DBS) is a surgical form of neuromodulation, and about 208,000 people have received it before 2021 (Davies, 2020). As demonstrated in the graphic, doctors make a small opening in the skull

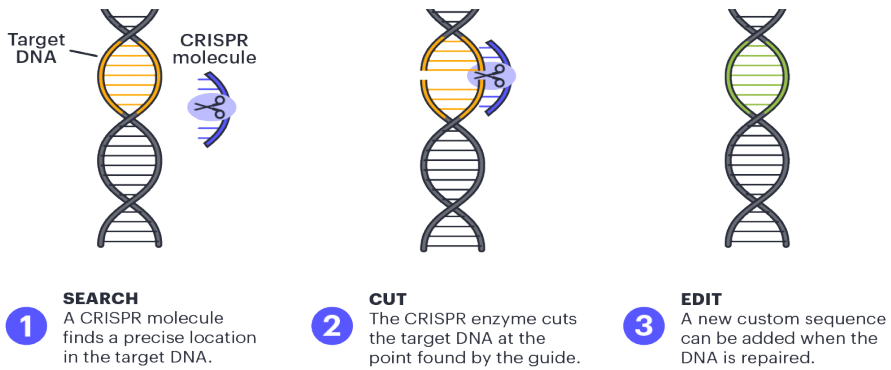


DBS is administered directly into the brain through an electrode (Holovina & Berezhnoi, 2017).

and insert an electrode, which is used to deliver the stimulant (Holovina & Berezhnoi, 2017). The electrode contains a pulse generator to manage the timing of the stimulant. DBS has been administered to patients suffering from a multitude of diseases, such as Parkingson disease, essential tremor, obsessive-compulsive disorder, and epilepsy.

Patients suffering from seizures who have received DBS showed an improvement in memory, and patients with epilepsy have shown improvement with visual memory (Davies, 2020). As the effects of neuromodulation prove to be beneficial to patients, doctors are now investigating DBS in the treatment of Alzheimer's disease. Specifically for Alzheimer's, DBS can induce an increase in cerebral glucose metabolism, stimulate hippocampal acetylcholine release, increase nerve growth factor, increasing neuronal and synaptic activity, and others. These are all essential for the brain to function as normal.

Gene editing and therapy will be able to benefit many people with neurological disorders by directly correcting pathogenic mechanisms, neuroprotection, neurorestoration, and symptom control (Sudhakar & Richardson, n.d.). The effectiveness of gene editing is dependent on the study of the disease under consideration. Scientists are hopeful that the availability of gene editing will help the development of therapeutics under development for neurological disorders.



Example of gene editing with the CRISPR enzyme (Doudna, n.d.).

Specifically for Alzheimer's disease, APOE, or the "Alzheimer's gene," is considered to be the leading risk factor for the disease (Prabhune, 2021). However, there are about 50 genes associated with Alzheimer's and related dementias. Recently, CRISPR, or Clustered Regularly Interspaced Short Palindromic Repeats, has been used to edit genes in brain cells to prevent Alzheimer's. As depicted in the graphic, CRISPR edits genes by cutting DNA and allowing the DNA to repair itself (Doudna, n.d.). By utilizing this method, CRISPR was found to reduce Alzheimer's likelihood by a factor of four (Prabhune, 2021). Specifically for Alzheimer's disease, APOE, or the "Alzheimer's gene," is considered to be the leading risk factor for the disease (Prabhune, 2021). However, there are about 50 genes associated with Alzheimer's and related dementias. Recently, CRISPR, or Clustered Regularly Interspaced Short

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The brain is the most complex structure in the human body, making it difficult to develop treatments for neurodegenerative diseases. Recently, however, the field of neuroscience has undergone a renaissance in its approach to the treatment and examination of these diseases. These novel approaches in developing treatments will prove beneficial as more research is done related to it.

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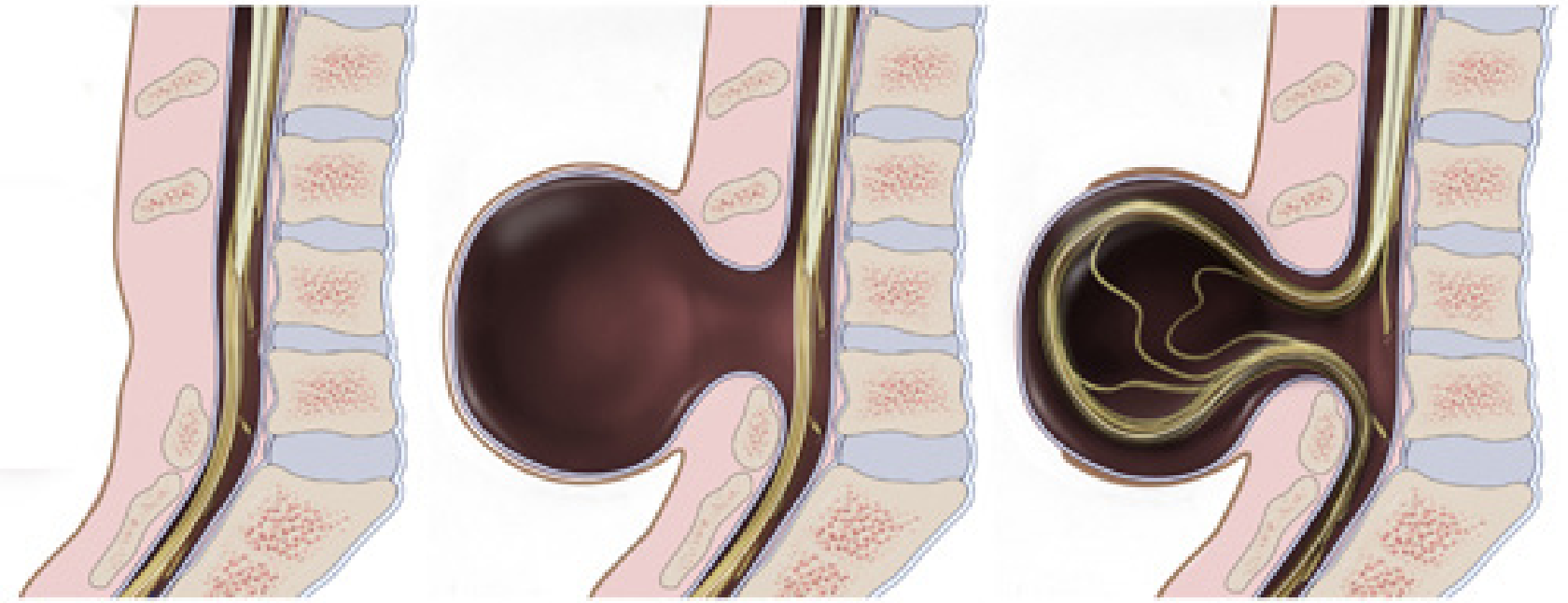
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7. A Promising "CuRe" for Spina Bifida

Leigh Foran, '23

Spina bifida is a condition that occurs when the neural tube, or the embryonic brain and spinal cord, does not close all the way (Munakomi et al., 2022). When this happens, the vertebrae and surrounding bone that protect the spinal cord do not close as they should, and therefore do not form properly. Oftentimes, this results in damage to the spinal cord and nerves ("What is Spina Bifida?", 2020).

To address this condition, which can cause walking and mobility problems, orthopedic complications, and accumulation of fluid in the brain ("Spina bifida", 2022), scientists at UC Davis Health have begun the CuRe Trial: Cellular Therapy for In Utero Repair of Myelomeningocele. This is the world's first spina bifida treatment utilizing stem cells, which are cells that have the ability to become many different types of cells throughout the body. They can even develop into specialized cells, such as muscle cells, blood cells, and brain cells ("Stem Cells", n.d.). These fascinating cells yield promising outlooks for the treatment of spina bifida.

Procuring the Stem Cells

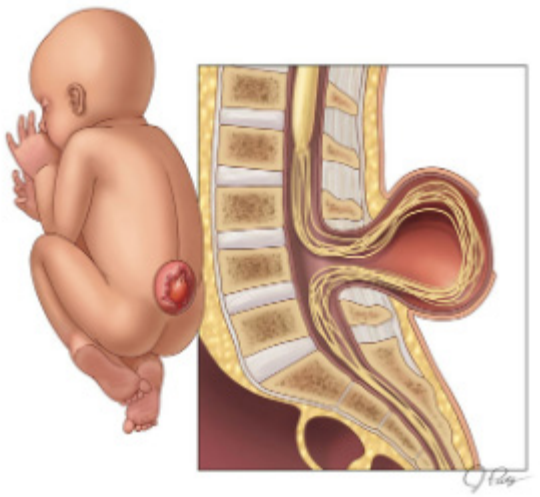
With the goal of minimizing paralysis and other abnormal defects as a result of spina bifida, the team at UC Davis utilized mesenchymal stem cells from placental tissue (Tomiyoshi, 2022). These are known to be some of the most promising types of stem cells in regenerative medicine. To fund the procurement of this vital asset, the university received a $9 million state grant from the California Institute for Regenerative Medicine (Tomiyoshi, 2022).

To procure these cells, the team of scientists and doctors specially engineered them in the UC Davis stem cell manufacturing facility according to FDA guidelines. These cells are known as the "paramedics" of the body because they create important substances that aid in cellular healing ("First stem cell clinical trial", 2021). At UC Davis's Institute for Regener-

ative Cures in Sacramento, the cells were produced and screened in a highly specialized laboratory. After this, the cells were assembled onto a special patch, tested for purity and sterility once again, and then hand-delivered to the fetal surgery operating rooms.

The Groundbreaking Surgeries

Performing the fetal surgery and stem cell procedure on each participant of the trial required a 40-person operating and cell preparation team. They worked to ensure that the procedure took place successfully while monitoring the health and safety of both the mother and fetus. During the procedure, stem cells were placed directly onto each fetus' spinal cord using the special patch (Farmer, 2022). This worked to repair the spina bifida defect. Myelomeningocele spina bifida results in a portion of the spinal cord being exposed in a sac through an opening in the spine. Through the surgeries, these protruding masses of spinal fluid were covered by the patches (Tomiyoshi, 2022).



Spina bifida's development in utero *(Farmer, 2022)*

The Results

Diana Farmer, the world's first female fetal surgeon, was the principal investigator of this study. Reporting on the results of the surgeries, she wrote, "Placement of the fetal patch went off without a hitch and mother and fetus did great!" The positive results of the procedures are proving to be an exciting option for babies with this condition.
Weeks after the surgery took place, the first babies receiving this groundbreaking fetal surgery were born. Mother Michelle Johnson and her baby Tobi are a happy example of this study's successful results.

"I am so thankful to be part of this journey to find a cure for spina bifida for Tobi and for so many others," Johnson said. "They are advancing medicine at UC Davis Health and Tobi is proof of that." Another mother, Emily, and her baby, Robbie, have greatly benefited from Dr. Farmer's work. Robbie, who was expected to be born with leg paralysis, was wiggling her toes just days after her delivery date. Emily said, "this experience has been larger than life and has exceeded every expectation. I hope this trial will enhance the quality of life for so many patients to come."

Though the CuRe team is hesitant to draw definitive conclusions, the work at UC Davis

Health indicates an exciting opportunity to improve the quality of life of thousands of babies throughout the world.

Looking Ahead

Although the results are promising, the babies in this trial will be monitored by the research team for 30 months after they are born ("First stem cell clinical trial", 2021). This is a crucial next step to ensure the stem cell surgical procedure's safety and efficacy.

"A successful treatment for MMC would relieve the tremendous emotional and economic cost burden on families," Farmer stated. She hopes her groundbreaking treatment can make up for "all the pain and suffering, specialized childcare, and lost time for unpaid caregivers such as parents." This potential cure serves as a bright future for thousands of families across the U.S.



Emily with her baby Robbie (Tomiyoshi, 2022)

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8. The Physical and Psychological Impacts of Music on Animals

Isabel Jo, ‘26

Music and other sounds stimulate nearly all parts of the human brain. It is widely accepted that the frontal lobe is the most fundamental part of the human brain. By listening to music, humans are capable of improving time-perceiving abilities, enhancing intelligence, strengthening their immune systems and more through the frontal lobe (Suguya and Yonetani). The temporal lobes, which sit on the sides of the brain, are responsible for allowing humans to appreciate the music they hear. The right hemisphere of the temporal lobe interprets music and sounds. The Wernicke’s area of the brain, located in the left hemisphere, is responsible for comprehending both written and spoken language. It is used to enjoy and analyze music. The nucleus accumbens of the brain releases dopamine to the body. Much like a drug, music can increase dopamine production. This is why many people become “addicted” to music and want to hear songs repetitively. The amygdala is the part of your brain responsible for triggering and processing emotions. Music significantly impacts the amygdala, controlling your “fight or flight” instincts. The hypothalamus is the part of your brain that controls homeostasis, regulating chemicals and hormones that control things such as heart rate, body temperature and mood (Suguya and Yonetani). Music impacts all of these parts of your brain, which work together to respond to auditory stimulus and impact other parts of your body.

A study by Wright State University found that music can significantly impact people’s heart rates, after testing 24 high school students. The student’s heart rates were taken through heart monitors while they listened to six different genres of music. On average, the subject’s heart rates demonstrated a significant increase when they listened to rock music; contrastingly, their heart rates experienced a notable decrease when listening to classical

music (Todd, 2015).

Listening to music has also been linked to a decrease in anxiety, an increase in prosocial behaviors, and a decrease in blood pressure (Kogan et Al, 2012). While the sample size for this experiment was relatively small, the emerging patterns suggest that not only can music affect people's emotions, but that different types of music can evoke physical changes in humans.

How does music impact dogs?

A study conducted in 2012 discovered the impacts of different genres of music on kenneled dogs. Dogs living in kennels are often subject to severe anxiety, also known as "kennel stress" (Pomfret, 2021). Hence, researchers Lori R. Kogan and Allen A. Simon aimed to discover whether certain types of music would be able to aid these dogs' nervousness. The experiment consisted of 117 dogs and included both rescue dogs and boarded dogs. The dogs listened to three types of music: 4 selections of classical music, with an average tempo of 121 BPM (beats per minute), 3 selections of heavy metal, with an average 131 BPM, and one selection of psychological music specifically composed for the purpose of dog relaxation. The dogs were exposed to each condition for 45 minutes at a time, with each period followed by 15 minutes of silence. With classical music playing, the dogs slept for, on average, between 3.7 and 6 percent of the time. This is a significant increase from the percentage of time they spent sleeping while listening to heavy metal music, between 0.8 and 1.2 percent. Furthermore, the dogs spent an average of 0.9 to 2.8 percent of the time shaking (unconsciously making small, fast movements—a symptom of anxiety) while classical music played; by contrast, between 37.8 and 71.2 percent of the time was spent shaking while listening to heavy metal music. For the period during which the psychological music was playing, the researchers found that the dogs spent 1.4 percent of the time sleeping and 0.5 percent of the time shaking. Overall, dogs demonstrated a calm response to both the classical and psychological music and experienced increased symptoms of anxiety when listening to heavy metal (Kogan et Al, 2012). Therefore, the conclusion can be made that playing tranquil music for dogs in kennels can help alleviate their anxiety. This strongly suggests that similar music could be beneficial to dogs in other stressful environments, such as a veterinary office or when traveling.



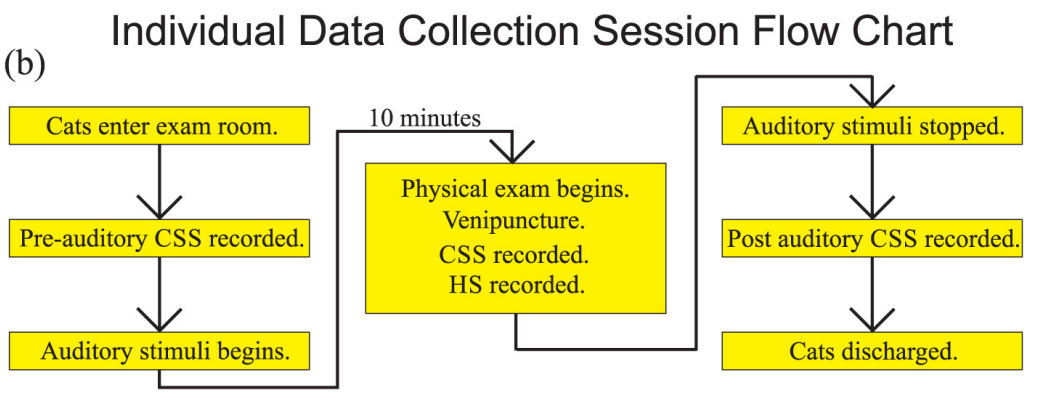
[Golden Retriever Wearing Headphones], Shutterstock

How does music impact rats?

An organism's central nervous system can be significantly altered by its environment at a young age. In fact, the development of functionality and processing abilities of a rodent's auditory cortex is largely dependent on the animal's initial acoustic environment. An enriching environment in which a rat is exposed to aural stimulation has been shown to improve the sensitivity, latency and response strength of the rat's auditory cortex neurons (Zu, Yu, Zhang, and Sun, 2008). Furthermore, researchers have found that exposure to classical music recordings, such as Mozart, has been able to assist hypertensive rats by lowering their blood pressure (Bryant, 2013). This is a similar result to that of the canine's exposure to music. In the case of the rodents, it was found that this blood pressure reducing effect was due to the classical music containing high frequencies that were within the optimal range for a rat's hearing sensitivity (4k-16kHz) (Bryant, 2013). Overall, exposing rodents to music from a young age has been proven to benefit their health and intelligence.

How does music impact cats?

Scientists have utilized human music research to learn more about how cats may respond to aural stimulation. The use of music has been an increasingly popular strategy in human medicine, and has been directly linked to decreasing anxiety symptoms. Furthermore, research has demonstrated that the types of music with the most significant benefit to humans are those with beats similar to a human's resting heart rate and frequencies within the human vocal range (News Staff, 2022). Researchers Amanda Hampton, Alexandra Ford, Roy E Cox, Chin-Chi Liu, and Ronald Koh conducted a study in which they explored whether similar benefits could be provided to domestic cats by producing cat-specific music, designed specifically to please feline animals. They exposed the cats to classical music, since that genre of music proved to be the most beneficial to homosapiens. During three auditory stimuli tests, each of which were 2 weeks apart, the cats' stress scores (CSS) were recorded in response to silence, classical music, and cat-specific music. The testing process can be seen in figure B. Results demonstrated that the CSS were very similar following both the classical music and silence. However, after exposure to the music specifically designed for the cats, the overall CSS were significantly lower than previously (Hamp-



(Hampton, 2019)

ton et al., 2019). The conclusion can be made that playing cat-specific music in veterinary clinics or other stressful environments would benefit the well being of domestic cats, thus making life easier for many veterinarians and cat owners.

Summary

Overall, research has made it clear that music can enhance the lives of many species in addition to humans. Both classical music and species-specific music have been directly linked to positive changes in dogs, rodents and cats; all animals that are commonly found as house pets. By lowering heart rates, decreasing blood pressure and reducing symptoms of anxiety, music has been proven to be a useful tool towards improving the wellbeing of animals in both veterinary and household settings.

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9. A Different Side of the Water Crisis

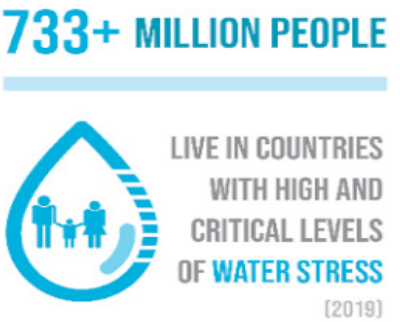
Olivia Cohn, '26

Water is a vital part of society: every day, it is used for sewage systems, consumption, medical care, renewable energy, and more. However, over 700 million people across the globe lack access to this necessity (Bostrom, 2015). This issue especially impacts women and girls who are often tasked with water collection. Women are responsible for 72% of the water collected in Sub-Saharan Africa (Corwin, 2022). They spend hours, multiple times per day, waiting in long lines at community water kiosks or walking to distant sources like rivers and ponds. This is time away from attending school or earning income. Most of the time, the water they worked so hard to collect is unsafe to drink. Diseases from dirty water kill more people annually than all forms of violence, including war (Charity: Water, 2022). Time spent collecting this diseased water or seeking a safe place to go accounts for billions of dollars in lost economic opportunities.



(United Nations, 2022)

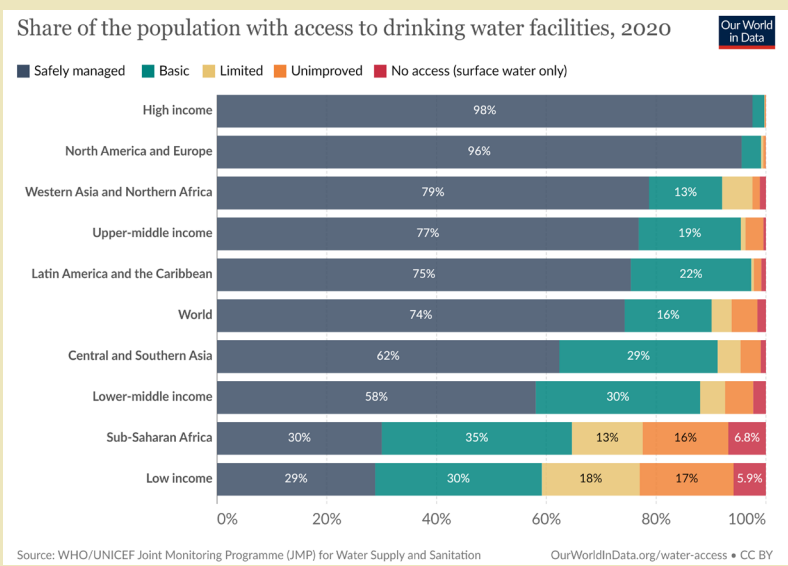
However, with clean water, everything changes, and we can unlock economic development for many who today do not have access. Better access to water would support keeping kids in school, especially girls, as they would spend more time in class and less time gathering basic human necessities. Furthermore, clean water and proper toilets at school would mean teenage girls would not have to stay home for a week out of every month. Giving girls the opportunity to stay in school and commit more time to learning allows them to start businesses, improve their homes, and take charg e of their futures. Solving the water crisis is not just about allowing communities to survive, but empowering girls to reach their full potential.



(United Nations, 2022)

To solve this, we must turn towards technology. Recycling water is a prominent solution being utilized. Wastewater reuse has existed for thousands of years; all water is naturally recycled and reused as part of the hydrologic cycle. But with new technology and systems, we are advancing this concept on a larger scale. Water reuse enhances water security, sustainability, and resilience, bolstering local water supplies, improving water quality, saving energy, and reducing wastewater disposal costs.

The COVID-19 pandemic has increased awareness of the water crisis's extent and consequences. Water access can be linked to everything; water is a vital part of medicare, necessary in infrastructure, required for food, and crucial for farming. The result of these problems intertwining leads to girls missing hours of school every week to gather this essential recourse. As the World Bank says, access to water supply and sanitation gaps are among the greatest risks to economic progress, poverty eradication, and sustainable development. Utilizing water reuse and recycling will help so many girls reach their potential and create a future in which they can thrive.



(Ritchie and Roser, 2021)

Green technology and the tools of science are powerful, even more so when we combine these tools with

ethics and civics. The effects of water insecurity do not recognize national borders. Similarly, our approach to creating solutions must harness the best technology and the most diverse talents and innovators worldwide. Technology, science, as well as empathy, and commitment to our communities will help us improve access to clean water. This issue is disproportionately limiting and challenging girls. Taking a step forward in reducing gender inequalities calls for a closer look into the water crisis.

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10. Forever Chemicals, a Forever Problem

Paige Foran, ‘26

Scientists have established that precipitation is a major, vital component of every single environment; ecosystems would quite literally fail without it. But what happens when it becomes dangerous? The medical world has recently encountered this very question. Specifically, the presence of “forever chemicals” has become a pressing concern, rendering rainwater unsafe to drink globally. These human-caused chemicals can last for over thousands of years, putting organisms all over the world in a hole that advances it may be impossible to dig ourselves out of.

Hence, these “forever chemicals,” which are also known as polyfluoroalkyl substances (PFAS), are chemicals from man-made products including water-resistant clothes, food packaging, nonstick cooking materials, and more. Some of the most frequently studied of these contaminants are perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS). While the use of both of these have been discontinued in the United States, these chemicals still remain legal in other areas of the world; as a result, if a global effort isn’t made, individual action will have very little effect. These chemicals simply do not break down (*What are PFAS?, 2022*). In order to stop global status from worsening, let alone reverse its progress, the production of these must be stopped altogether.

PFAS released from factories can exit manufacturing locations via gaseous emissions, by which they deposit themselves in nearby bodies of water. Landfill overflow can also land in these bodies of water. This water, often taken for farmwork, is treated by certain wastewater treatment plants, known as “biosolids,” whose outputs can include PFA-infested sludge. This sludge is often used as fertilizer on farmlands, damaging the crops by causing molecular discrepancies and infesting groundwater with PFAS. Such groundwater is evaporated and condensed, so when it comes down as precipitation, it contains PFAS and the cycle

begins again. (Puckett, n.d.)

Scientists have found significant increases of these substances in many areas, including rainwater, soil, snow, and, horrifyingly, human blood. While it may be challenging to specifically link environmental alterations to health effects, members of the scientific community have been able to reasonably conclude that excessive exposure to PFAS can lead to issues such as infertility, pregnancy problems, cancer, learning and behavioral issues in children, heightened cholesterol, and mitigations of the immune system's function (Crist, 2022). Naturally, with the newfound knowledge of the toxicity of these chemicals, countries have been decreasing the maximum content of PFAS in water that has been cleared to ingest. Ian Cousins, PhD, a renowned study leader and professor of environmental science at Stockholm University, explicitly said, "Based on the latest U.S. guidelines for PFOA in drinking water, rainwater everywhere would be judged unsafe to drink. Although in the industrial world we don't often drink rainwater, many people around the world expect it to be safe to drink and it supplies many of our drinking water sources" (*It's Raining*, 2022). He and his colleagues, after measuring the content of the 4 main types of perfluoroalkyl acids in the atmosphere (perfluorooctanesulfonic acid, perfluorooctanoic acid, perfluorohexane sulfonic acid, and perfluorononanoic acid) have determined that the amount of these contaminants in the environment are far too high in relation to U.S. guidelines and also many limits for areas across Europe (Crist, 2022).

Many countries rely on rainwater as the main source of water for their economy, and PFAS-contamination can seriously harm their nationwide health. For example, countries like Brazil have implemented a new system called Rooftop Rainwater Harvesting (RTRWH) in which precipitation is collected on the rooftops of buildings and purified for reuse (*Rainwater harvesting*, 2015). However, these purification methods are not accustomed to filtering out PFAS; therefore, if humans drink this purified water, they will still be ingesting PFAS.

With the abundance of these chemicals, it has become significantly more dangerous to ingest rainwater anywhere; of course, a mere drop will not be enough to spur a fatal condition or disease in the body. However, without proper awareness of the situation in developing and third-world countries, members of these communities are at severe risk to develop the potential health problems associated with these PFAS (Puckett, n.d.). Furthermore, studies have shown that higher levels of certain PFAS can lead to heightened risk for breast and kidney cancer, high cholesterol, slight decreases in infant weight, and more. If such PFAS were consumed nationwide, these countries could land in a massive healthcare deficit and lose funding for other projects (*What are the Health*, 2022).

But is there a solution to such a seemingly debilitating contamination? As of now, the best course of action is to stop chemical pollution, which is significantly easier said than done. If humanity ceases to contaminate nature and its mediums, the future will be secured; this

water, when evaporated, will return as nontoxic precipitation. However, as of right now, there is nothing we can do to change the toxicity of today's rainwater, as the chemicals have already been absorbed.

All in all, the toxicity of precipitation is a global issue that cannot go unaddressed. Not only will it become more and more debilitating as time goes on, but it is yet another environmental factor that humans must deal with in addition to global warming, species extinction, and more. So, the next time you open your mouth to catch a few raindrops, think twice about the forever impacts possible from these forever chemicals.

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Header Image:

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